

MULTILASER^{PRO))}

MANUAL DE CONFIGURAÇÃO

Série TITAN – C600, C610, C620 e C650

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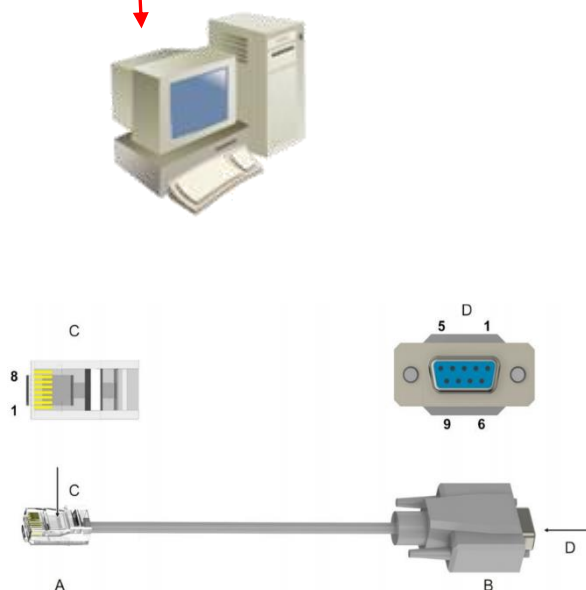
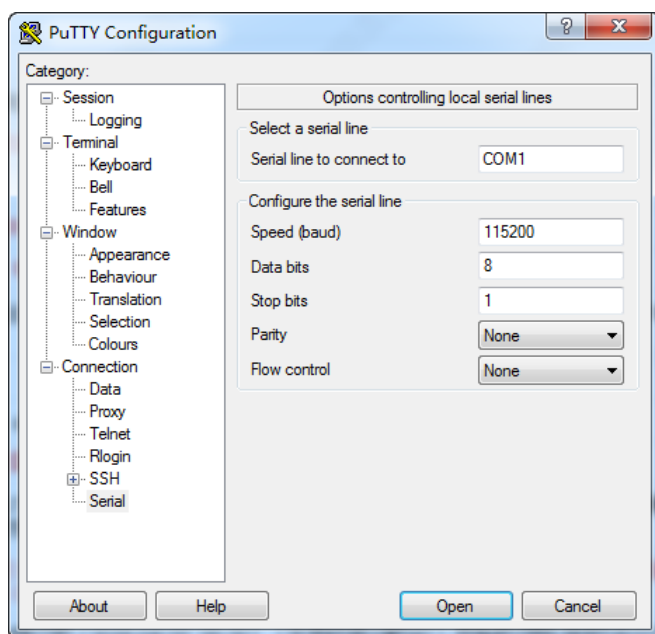
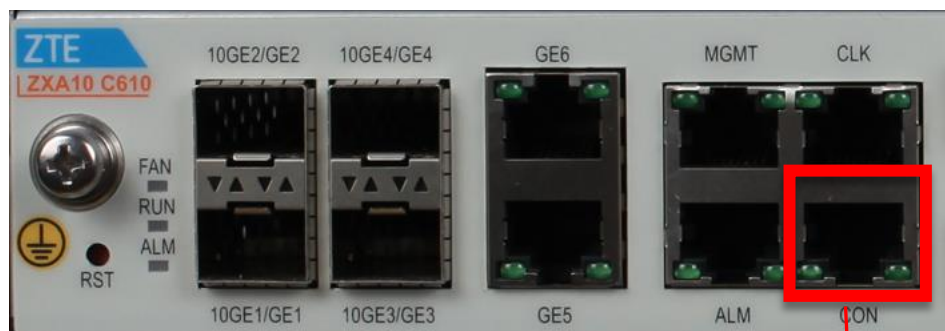
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1. ATIVAÇÃO OLT

1.1. Acesso ao Equipamento

É necessário utilizar o cabo fornecido junto com o equipamento, pois seu arranje de pinos é diferente do padrão de outros equipamentos. O acesso via cabo console pode ser feito através da porta CLI presente na controladora da OLT. (Speed 115200) através de um software terminal.



1.2. Configurar Data e Hora

OLT#clock set 09:35:00 12-22-2021

```
OLT#clock set 09:35:00 12-22-2021
OLT#
```

1.3. Backup

1.3.1. Backup Manual

```
OLT#  
OLT#copy ftp root: /datadisk0/DATA0/startrun.dat //10.0.250.253/startrun.dat@zte:zte  
Connect successfully! Start copying file  
100% completed 00:00:00  
Put file successfully! Sent 32065 bytes!  
OLT#
```

```
OLT#copy ftp root: /datadisk0/DATA0/startrun.dat //10.0.250.253/startrun.dat@zte:zte  
Connect successfully! Start copying file  
100% completed 00:00:00  
Put file successfully! Sent 32065 bytes!  
OLT#
```

1.3.2. Backup Automático

```
OLT#conf t  
Enter configuration commands, one per line. End with CTRL/Z.  
OLT(config)#startrun backup-to server 10.0.250.253 transport-type ftp username backup  
password zte path ZTE  
OLT(config)#auto-write everyday 11:10:00  
OLT(config)#
```

```
OLT#conf t  
Enter configuration commands, one per line. End with CTRL/Z.  
OLT(config)#startrun backup-to server 10.0.250.253 transport-type ftp username backup password zte path ZTE  
OLT(config)#auto-write everyday 11:10:00  
OLT(config)#  
OLT#
```

Obs.: Para ter sucesso nos dois cenários de backup, você necessita ter acesso a um Servidor de FTP local ou remoto.

1.4. Alterar Hostname da OLT

```
OLT#conf t  
Enter configuration commands, one per line. End with CTRL/Z.  
OLT(config)#hostname OLT2  
OLT2(config)#
```

```
OLT#conf t  
Enter configuration commands, one per line. End with CTRL/Z.  
OLT(config)#hostname OLT2  
OLT2(config)#
```

1.5. Habilitar Acesso Externo

1.5.1. Telnet

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#line telnet server enable

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#line telnet server enable
```

1.5.2. SSH

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#ssh server enable

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#ssh server enable
```

1.6. Criação de Usuário

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#system-user

ZXAN(config-system-user)#user-name zte

ZXAN(config-system-user-username)#password Abc@101231!

ZXAN(config-system-user-username)#bind authentication-template 1

ZXAN(config-system-user-username)#bind authorization-template 1

ZXAN(config-system-user-username)#exit

ZXAN(config-system-user)#exit

ZXAN(config)#aaa-authentication-template 2001

ZXAN(config-aaa-authen-template)#aaa-authentication-type local

ZXAN(config-aaa-authen-template)#exit

ZXAN(config)#aaa-authorization-template 2001

ZXAN(config-aaa-author-template)#aaa-authorization-type local

ZXAN(config-aaa-author-template)#exit

ZXAN(config)#system-user

ZXAN(config-system-user)#authentication-template 1

ZXAN(config-system-user-authen-temp)#bind aaa-authentication-template 2001

ZXAN(config-system-user-authen-temp)#exit

ZXAN(config-system-user)#authorization-template 1

ZXAN(config-system-user-author-temp)#bind aaa-authorization-template 2001

ZXAN(config-system-user-author-temp)#local-privilege-level 15

ZXAN(config-system-user-author-temp)#exit

ZXAN(config-system-user)#

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#system-user
ZXAN(config-system-user)#user-name zte
ZXAN(config-system-user-username)#password Abc@101231!
ZXAN(config-system-user-username)#bind authentication-template 1
ZXAN(config-system-user-username)#bind authorization-template 1
ZXAN(config-system-user-username)#exit
ZXAN(config-system-user)#exit
ZXAN(config)#aaa-authentication-template 2001
ZXAN(config-aaa-authen-template)#aaa-authentication-type local
ZXAN(config-aaa-authen-template)#exit
ZXAN(config)#aaa-authorization-template 2001
ZXAN(config-aaa-author-template)#aaa-authorization-type local
ZXAN(config-aaa-author-template)#exit
ZXAN(config)#system-user
ZXAN(config-system-user)#authentication-template 1
ZXAN(config-system-user-authen-temp)#bind aaa-authentication-template 2001
ZXAN(config-system-user-authen-temp)#exit
ZXAN(config-system-user)#authorization-template 1
ZXAN(config-system-user-author-temp)#bind aaa-authorization-template 2001
ZXAN(config-system-user-author-temp)#local-privilege-level 15
ZXAN(config-system-user-author-temp)#exit
ZXAN(config-system-user)#
ZXAN#
```

1.7. Habilitar Portas PON

```
interface gpon_olt-1/3/1
no shutdown
exit
```

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#interface gpon_olt-1/3/1
ZXAN(config-if-gpon_olt-1/3/1)#no shutdown
```

Obs.: Não é possível liberar todas as portas de uma só vez como na série anterior.

1.8. Criação de VLAN

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#vlan 999

ZXAN(config-sub-999)#description TREINAMENTO

ZXAN(config-sub-999)#

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#vlan 999
ZXAN(config-sub-999)#description TREINAMENTO
ZXAN(config-sub-999)#
```

1.9. Criação de Perfil de VLAN

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#gpon

ZXAN(config-gpon)#onu profile vlan 999 tag-mode tag cvlan 999

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile vlan 999 tag-mode tag cvlan 999
ZXAN(config-gpon)#
```

1.10. Habilitar Portas de UPLINK

Utilizando o comando [3.3.8](#) você seleciona a porta que deseja utilizar e segue o exemplo abaixo:

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#interface xgei-1/1/1

ZXAN(config-if-xgei-1/1/1)#no shutdown

ZXAN(config-if-xgei-1/1/1)#

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#interface xgei-1/1/1
ZXAN(config-if-xgei-1/1/1)#no shutdown
ZXAN(config-if-xgei-1/1/1)#
```

1.11. Adicionar VLAN's a UPLINK

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#interface xgei-1/1/1

ZXAN(config-if-xgei-1/1/1)#switchport vlan 99 tag

ZXAN(config-if-xgei-1/1/1)#

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#interface xgei-1/1/1
ZXAN(config-if-xgei-1/1/1)#switchport vlan 99 tag
ZXAN(config-if-xgei-1/1/1)#
```

1.12. Gerência INBAND e OUTBAND

INBAND:

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#interface vlan999

ZXAN(config-if-vlan999)#ip address 14.14.14.14 255.255.255.0

ZXAN(config-if-vlan999)#

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#interface vlan999
ZXAN(config-if-vlan999)#ip address 14.14.14.14 255.255.255.0
ZXAN(config-if-vlan999)#
```

OUTBAND:

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#interface mgmt_eth

ZXAN(config-if-mgmt_eth)#ip address 14.14.14.2 255.255.255.0

ZXAN(config-if-mgmt_eth)#exit

ZXAN(config)#ip route vrf mng 0.0.0.0 0.0.0.0 14.14.14.1

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#interface mgmt_eth
ZXAN(config-if-mgmt_eth)#ip address 14.14.14.2 255.255.255.0
ZXAN(config-if-mgmt_eth)#exit
ZXAN(config)#ip route vrf mng 0.0.0.0 0.0.0.0 14.14.14.1
ZXAN(config)#
```

2. CONFIGURAÇÃO ONU

2.1. Perfil de ONU

2.1.1. F601



```
ZXAN#conf t
```

Enter configuration commands, one per line. End with CTRL/Z.

```
ZXAN(config)#pon
```

```
ZXAN(config-pon)#$description 1GE_BRIDGE max-tcont 7 max-gemport 32 max-switch-perslot 8 max-flow-perswitch 8 max-veip 4
```

```
ZXAN(config-pon)# onu-type-if F601 eth_0/1
```

```
ZXAN(config-pon)#
```

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#pon
ZXAN(config-pon)#$description 1GE_BRIDGE max-tcont 7 max-gemport 32 max-switch-perslot 8 max-flow-perswitch 8 max-veip 4
ZXAN(config-pon)# onu-type-if F601 eth_0/1
ZXAN(config-pon)#
```

2.1.2. F612



```
ZXAN#conf t
```

Enter configuration commands, one per line. End with CTRL/Z.

```
ZXAN(config)#pon
```

```
ZXAN(config-pon)#$oip max-tcont 7 max-gemport 32 max-switch-perslot 8 max-flow-
perswitch 8 max-iphost 5 max-ipv6-host 5
ZXAN(config-pon)# onu-type-if ZTE-F612 eth_0/1
ZXAN(config-pon)# onu-type-if ZTE-F612 eth_0/2
ZXAN(config-pon)# onu-type-if ZTE-F612 pots_0/1
```

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#pon
ZXAN(config-pon)#$oip max-tcont 7 max-gemport 32 max-switch-perslot 8 max-flow-perswitch 8 max-iphost 5 max-ipv6-host 5
ZXAN(config-pon)# onu-type-if ZTE-F612 eth_0/1
ZXAN(config-pon)# onu-type-if ZTE-F612 eth_0/2
ZXAN(config-pon)# onu-type-if ZTE-F612 pots_0/1
```

2.1.3. F660



```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#pon
ZXAN(config-pon)# onu-type ZTE-F660 gpon max-tcont 5 max-gemport 32 max-switch-
perslot 2 max-flow-perswitch 8 max-iphost 5
ZXAN(config-pon)# onu-type-if F660 eth_0/1
ZXAN(config-pon)# onu-type-if F660 eth_0/2
ZXAN(config-pon)# onu-type-if F660 eth_0/3
ZXAN(config-pon)# onu-type-if F660 eth_0/4
ZXAN(config-pon)# onu-type-if F660 pots_0/1
ZXAN(config-pon)# onu-type-if F660 pots_0/2
ZXAN(config-pon)# onu-type-if F660 wifi_0/1
ZXAN(config-pon)# onu-type-if F660 wifi_0/2
ZXAN(config-pon)# onu-type-if F660 wifi_0/3
ZXAN(config-pon)# onu-type-if F660 wifi_0/4
```

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#pon
ZXAN(config-pon)# onu-type ZTE-F660 gpon max-tcont 5 max-gemport 32 max-switch-perslot 2 max-flow-perswitch 8 max-iphost 5
ZXAN(config-pon)# onu-type-if F660 eth_0/1
ZXAN(config-pon)# onu-type-if F660 eth_0/2
ZXAN(config-pon)# onu-type-if F660 eth_0/3
ZXAN(config-pon)# onu-type-if F660 eth_0/4
ZXAN(config-pon)# onu-type-if F660 pots_0/1
ZXAN(config-pon)# onu-type-if F660 pots_0/2
ZXAN(config-pon)# onu-type-if F660 wifi_0/1
ZXAN(config-pon)# onu-type-if F660 wifi_0/2
ZXAN(config-pon)# onu-type-if F660 wifi_0/3
ZXAN(config-pon)# onu-type-if F660 wifi_0/4
ZXAN(config-pon)#
```

2.1.4. F670L



ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#pon

ZXAN(config-pon)#\$,4WiFi5 max-tcont 7 max-gemport 32 max-switch-perslot 8 max-flow-perswitch 8 max-iphost 5 max-ipv6-host 5

ZXAN(config-pon)# onu-type-if F670L eth_0/1

ZXAN(config-pon)# onu-type-if F670L eth_0/2

ZXAN(config-pon)# onu-type-if F670L eth_0/3

ZXAN(config-pon)# onu-type-if F670L eth_0/4

ZXAN(config-pon)# onu-type-if F670L pots_0/1

ZXAN(config-pon)# onu-type-if F670L wifi_0/1

ZXAN(config-pon)# onu-type-if F670L wifi_0/2

ZXAN(config-pon)# onu-type-if F670L wifi_0/3

ZXAN(config-pon)# onu-type-if F670L wifi_0/4

ZXAN(config-pon)# onu-type-if F670L wifi_0/5

ZXAN(config-pon)# onu-type-if F670L wifi_0/6

ZXAN(config-pon)# onu-type-if F670L wifi_0/7

ZXAN(config-pon)# onu-type-if F670L wifi_0/8

ZXAN(config-pon)#

```
ZXAN#conf t
```

```
Enter configuration commands, one per line. End with CTRL/Z.
```

```
ZXAN(config)#pon
```

```
ZXAN(config-pon)#$,4WiFi5 max-tcont 7 max-gemport 32 max-switch-perslot 8 max-flow-perswitch 8 max-iphost 5 max-ipv6-host 5
```

```
ZXAN(config-pon)# onu-type-if F670L eth_0/1
```

```
ZXAN(config-pon)# onu-type-if F670L eth_0/2
```

```
ZXAN(config-pon)# onu-type-if F670L eth_0/3
```

```
ZXAN(config-pon)# onu-type-if F670L eth_0/4
```

```
ZXAN(config-pon)# onu-type-if F670L pots_0/1
```

```
ZXAN(config-pon)# onu-type-if F670L wifi_0/1
```

```
ZXAN(config-pon)# onu-type-if F670L wifi_0/2
```

```
ZXAN(config-pon)# onu-type-if F670L wifi_0/3
```

```
ZXAN(config-pon)# onu-type-if F670L wifi_0/4
```

```
ZXAN(config-pon)# onu-type-if F670L wifi_0/5
```

```
ZXAN(config-pon)# onu-type-if F670L wifi_0/6
```

```
ZXAN(config-pon)# onu-type-if F670L wifi_0/7
```

```
ZXAN(config-pon)# onu-type-if F670L wifi_0/8
```

```
ZXAN(config-pon)#
```

2.1.5. F680



ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#pon

ZXAN(config-pon)#\$4WiFi5 max-tcont 7 max-gemport 32 max-switch-perslot 8 max-flow-perswitch 8 max-iphost 5 max-ipv6-host 5

ZXAN(config-pon)# onu-type-if F680 eth_0/1

ZXAN(config-pon)# onu-type-if F680 eth_0/2

ZXAN(config-pon)# onu-type-if F680 eth_0/3

ZXAN(config-pon)# onu-type-if F680 eth_0/4

ZXAN(config-pon)# onu-type-if F680 pots_0/2

ZXAN(config-pon)# onu-type-if F680 wifi_0/1

ZXAN(config-pon)# onu-type-if F680 wifi_0/2

ZXAN(config-pon)# onu-type-if F680 wifi_0/3

ZXAN(config-pon)# onu-type-if F680 wifi_0/4

ZXAN(config-pon)# onu-type-if F680 wifi_0/5

ZXAN(config-pon)# onu-type-if F680 wifi_0/6

ZXAN(config-pon)# onu-type-if F680 wifi_0/7

ZXAN(config-pon)# onu-type-if F680 wifi_0/8

ZXAN(config-pon)# onu-type-if F680 pots_0/1

ZXAN(config-pon)# onu-type-if F680 pots_0/2

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#pon

ZXAN(config-pon)#\$4WiFi5 max-tcont 7 max-gemport 32 max-switch-perslot 8 max-flow-perswitch 8 max-iphost 5 max-ipv6-host 5

ZXAN(config-pon)# onu-type-if F680 eth_0/1

ZXAN(config-pon)# onu-type-if F680 eth_0/2

ZXAN(config-pon)# onu-type-if F680 eth_0/3

ZXAN(config-pon)# onu-type-if F680 eth_0/4

ZXAN(config-pon)# onu-type-if F680 pots_0/2

ZXAN(config-pon)# onu-type-if F680 wifi_0/1

ZXAN(config-pon)# onu-type-if F680 wifi_0/2

ZXAN(config-pon)# onu-type-if F680 wifi_0/3

ZXAN(config-pon)# onu-type-if F680 wifi_0/4

ZXAN(config-pon)# onu-type-if F680 wifi_0/5

ZXAN(config-pon)# onu-type-if F680 wifi_0/6

ZXAN(config-pon)# onu-type-if F680 wifi_0/7

ZXAN(config-pon)# onu-type-if F680 wifi_0/8

ZXAN(config-pon)# onu-type-if F680 pots_0/1

ZXAN(config-pon)# onu-type-if F680 pots_0/2

ZXAN(config-pon)#

2.1.6. F6600



ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#pon

ZXAN(config-pon)#\$ax-tcont 7 max-gemport 32 max-switch-perslot 8 max-flow-perswitch 8 max-iphost 5 max-ipv6-host 5

ZXAN(config-pon)# onu-type-if F6600 eth_0/1

ZXAN(config-pon)# onu-type-if F6600 eth_0/2

ZXAN(config-pon)# onu-type-if F6600 eth_0/3

ZXAN(config-pon)# onu-type-if F6600 eth_0/4

ZXAN(config-pon)# onu-type-if F6600 pots_0/1

ZXAN(config-pon)# onu-type-if F6600 pots_0/2

ZXAN(config-pon)# onu-type-if F6600 wifi_0/1

ZXAN(config-pon)# onu-type-if F6600 wifi_0/2

ZXAN(config-pon)# onu-type-if F6600 wifi_0/3

ZXAN(config-pon)# onu-type-if F6600 wifi_0/4

ZXAN(config-pon)# onu-type-if F6600 wifi_0/5

ZXAN(config-pon)# onu-type-if F6600 wifi_0/6

ZXAN(config-pon)# onu-type-if F6600 wifi_0/7

ZXAN(config-pon)# onu-type-if F6600 wifi_0/8

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#pon

ZXAN(config-pon)#\$ax-tcont 7 max-gemport 32 max-switch-perslot 8 max-flow-perswitch 8 max-iphost 5 max-ipv6-host 5

ZXAN(config-pon)# onu-type-if F6600 eth_0/1

ZXAN(config-pon)# onu-type-if F6600 eth_0/2

ZXAN(config-pon)# onu-type-if F6600 eth_0/3

ZXAN(config-pon)# onu-type-if F6600 eth_0/4

ZXAN(config-pon)# onu-type-if F6600 pots_0/1

ZXAN(config-pon)# onu-type-if F6600 pots_0/2

ZXAN(config-pon)# onu-type-if F6600 wifi_0/1

ZXAN(config-pon)# onu-type-if F6600 wifi_0/2

ZXAN(config-pon)# onu-type-if F6600 wifi_0/3

ZXAN(config-pon)# onu-type-if F6600 wifi_0/4

ZXAN(config-pon)# onu-type-if F6600 wifi_0/5

ZXAN(config-pon)# onu-type-if F6600 wifi_0/6

ZXAN(config-pon)# onu-type-if F6600 wifi_0/7

ZXAN(config-pon)# onu-type-if F6600 wifi_0/8

ZXAN(config-pon)#

2.1.7. F8648p



ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#pon

ZXAN(config-pon)#\$cont 7 max-gemport 32 max-switch-perslot 8 max-flow-perswitch 8 max-iphost 5 max-ipv6-host 5

ZXAN(config-pon)#onu-type-if F8648P eth_0/1

ZXAN(config-pon)#onu-type-if F8648P eth_0/2

ZXAN(config-pon)#onu-type-if F8648P eth_0/3

ZXAN(config-pon)#onu-type-if F8648P eth_0/4

ZXAN(config-pon)#onu-type-if F8648P eth_0/5

ZXAN(config-pon)#onu-type-if F8648P pots_0/1

ZXAN(config-pon)#onu-type-if F8648P pots_0/2

ZXAN(config-pon)#onu-type-if F8648P wifi_0/1

ZXAN(config-pon)#onu-type-if F8648P wifi_0/2

ZXAN(config-pon)#onu-type-if F8648P wifi_0/3

ZXAN(config-pon)#onu-type-if F8648P wifi_0/4

ZXAN(config-pon)#onu-type-if F8648P wifi_0/5

ZXAN(config-pon)#onu-type-if F8648P wifi_0/6

ZXAN(config-pon)#onu-type-if F8648P wifi_0/7

ZXAN(config-pon)#onu-type-if F8648P wifi_0/8

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#pon

ZXAN(config-pon)#\$cont 7 max-gemport 32 max-switch-perslot 8 max-flow-perswitch 8 max-iphost 5 max-ipv6-host 5

ZXAN(config-pon)#onu-type-if F8648P eth_0/1

ZXAN(config-pon)#onu-type-if F8648P eth_0/2

ZXAN(config-pon)#onu-type-if F8648P eth_0/3

ZXAN(config-pon)#onu-type-if F8648P eth_0/4

ZXAN(config-pon)#onu-type-if F8648P eth_0/5

ZXAN(config-pon)#onu-type-if F8648P pots_0/1

ZXAN(config-pon)#onu-type-if F8648P pots_0/2

ZXAN(config-pon)#onu-type-if F8648P wifi_0/1

ZXAN(config-pon)#onu-type-if F8648P wifi_0/2

ZXAN(config-pon)#onu-type-if F8648P wifi_0/3

ZXAN(config-pon)#onu-type-if F8648P wifi_0/4

ZXAN(config-pon)#onu-type-if F8648P wifi_0/5

ZXAN(config-pon)#onu-type-if F8648P wifi_0/6

ZXAN(config-pon)#onu-type-if F8648P wifi_0/7

ZXAN(config-pon)#onu-type-if F8648P wifi_0/8

ZXAN(config-pon)#

ZXAN#~

2.2. Profile TCONT

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#gpon

ZXAN(config-gpon)#profile tcont 1G type 4 maximum 1024000

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#
ZXAN(config)#gpon
ZXAN(config-gpon)#profile tcont 1G type 4 maximum 1024000
ZXAN(config-gpon)#
ZXAN#
```

2.3. Profile Download

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#\$ traffic-profile 1G cir 1024000 cbs 1024 pir 1024000 pbs 1024 color-mode blind policer-type enhanced_mef coupling-flag enable

ZXAN(config)#

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#$r 1024000 cbs 1024 pir 1024000 pbs 1024 color-mode blind policer-type enhanced_mef coupling-flag enable
ZXAN(config)#
```

2.4. Provisionamento Manual

Se inicia pelo commando que pesquisa das onus não Provisionadas.

OLT#show pon onu unc

```
OLT#show pon onu unc
oltIndex      Model          SN              PW
-----
gpon_olt-1/1/1  F670LV9.0     ZTEGC4A8494B   GC4A8494B
```

Depois, acessar a interface da PON e Pesquisar qual ID está disponível para configurar.

OLT#conf t

Enter configuration commands, one per line. End with CTRL/Z.

OLT(config)#interface gpon_olt-1/1/1

OLT(config-if-gpon_olt-1/1/1)#show this

```
OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#inter
OLT(config)#interface gpo
OLT(config)#interface gpon_olt-1/1/1
OLT(config-if-gpon_olt-1/1/1)#show this
!<xpon>
onu 1 type F601 sn ZTEGCD7FF81F
onu 2 type F6600 sn ZTEGC7879B73
onu 11 type F670L sn ZTEGC4BA5555
onu 12 type F670L sn ZTEGC4BAE7A2
onu 20 type F660 sn ZTEGCB01EE7
onu 100 type F670L sn ZTEGCB01EE8
onu 101 type F670L sn ZTEGCB01EE9
bind-onu 12 profile line internet
!</xpon>
!<MSAN>
no shutdown
!</MSAN>
```

Com essas informações, adicionar a ONU.

OLT(config-if-gpon_olt-1/1/1)#onu 102 type F670L sn ZTEGC4A8494B

```
OLT(config-if-gpon_olt-1/1/1)#onu 102 ty
OLT(config-if-gpon_olt-1/1/1)#onu 102 type F670L sn
OLT(config-if-gpon_olt-1/1/1)#onu 102 type F670L sn ZTEGC4A8494B
OLT(config-if-gpon_olt-1/1/1)#
```

2.4.1. Configuração ONU

```
OLT(config)#interface gpon_onu-1/1/1:102
OLT(config-if-gpon_onu-1/1/1:102)#vport-mode manual
OLT(config-if-gpon_onu-1/1/1:102)#tcont 1 profile 1G
OLT(config-if-gpon_onu-1/1/1:102)#gemport 1 tcont 1
OLT(config-if-gpon_onu-1/1/1:102)#vport 1 map-type vlan
OLT(config-if-gpon_onu-1/1/1:102)#vport-map 1 1 vlan 1000
OLT(config-if-gpon_onu-1/1/1:102)#exit
```

```
OLT(config)#interface gpon_onu-1/1/1:102
OLT(config-if-gpon_onu-1/1/1:102)#show this
!<xpon>
vport-mode manual
tcont 1 profile 1G
gemport 1 tcont 1
vport 1 map-type vlan
vport-map 1 1 vlan 1000
!</xpon>
```

2.4.2. Configuração VPORT

OLT#conf t

Enter configuration commands, one per line. End with CTRL/Z.

OLT(config)#interface vport-1/1/1.102:1

OLT(config-if-vport-1/1/1.102:1)#service-port 1 user-vlan 1000 vlan 1000 ingress 1G egress 1G

OLT(config-if-vport-1/1/1.102:1)#

```
OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#interface vport-1/1/1.102:1
OLT(config-if-vport-1/1/1.102:1)#service-port 1 user-vlan 1000 vlan 1000 ingress 1G egress 1G
OLT(config-if-vport-1/1/1.102:1)#
```

Obs.: Caso não queira regular a sua conexão apenas pelo TCONT, não precisa informar o ingress e egress.

2.4.3. Configuração WAN em modo PPPOE

OLT(config)#pon-onu-mng gpon_onu-1/1/1:102

OLT(config-gpon-onu-mng)#service 1 gemport 1 vlan 1000

OLT(config-gpon-onu-mng)#wan-ip ipv4 mode pppoe username f601 password f601 vlan-profile 1000 host 1

OLT(config-if-vport-1/1/1.102:1)#exit

```
OLT(config)#pon-onu-mng gpon_onu-1/1/1:102
OLT(config-gpon-onu-mng)#show this
!<xpon>
  service 1 gemport 1 vlan 1000
  wan-ip ipv4 mode pppoe username f601 password ***** vlan-profile 1000 host 1
!</xpon>
```

2.4.4. Configuração WAN em modo DHCP

OLT(config)#pon-onu-mng gpon_onu-1/1/1:102

OLT(config-gpon-onu-mng)#service 1 gemport 1 vlan 1000

OLT(config-gpon-onu-mng)#wan-ip 1 ipv4 mode dhcp vlan-profile 1000 host 1

OLT(config-if-vport-1/1/1.102:1)#exit

```
OLT(config)#pon-onu-mng gpon_onu-1/1/1:102
OLT(config-gpon-onu-mng)#show this
!<xpon>
  service 1 gemport 1 vlan 1000
  wan-ip ipv4 mode dhcp vlan-profile 1000 host 1
!</xpon>
```

2.4.5. Configuração Bridge

OLT(config)#pon-onu-mng gpon_onu-1/1/1:102

OLT(config-gpon-onu-mng)#vlan port eth_0/1 mode tag vlan 1000

OLT(config-if-vport-1/1/1.102:1)#exit

```
OLT(config)#pon-onu-mng gpon_onu-1/1/1:102
OLT(config-gpon-onu-mng)#show this
!<xpon>
  service 1 gport 1 vlan 1000
  vlan port eth_0/1 mode tag vlan 1000
```

2.5. Acesso Remoto

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#pon-onu-mng gpon_onu-1/13/1:4

ZXAN(config-gpon-onu-mng)#security-mgmt 1 state enable mode forward ingress-type
iphost 1 protocol web

ZXAN(config-gpon-onu-mng)#

```
ZXAN(config)#
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#pon-onu-mng gpon_onu-1/13/1:4
ZXAN(config-gpon-onu-mng)#security-mgmt 1 state enable mode forward ingress-type iphost 1 protocol web
ZXAN(config-gpon-onu-mng)#
ZXAN#
```

2.6. WIFI

OLT#conf t

Enter configuration commands, one per line. End with CTRL/Z.

OLT(config)#pon-onu-mng gpon_onu-1/1/1:1

OLT(config-gpon-onu-mng)#ssid ctrl wifi_0/1 name TREINAMENTO

OLT(config-gpon-onu-mng)#ssid auth wpa wifi_0/1 key 12345678

OLT(config-gpon-onu-mng)#ssid ctrl wifi_0/5 name TREINAMENTO5

OLT(config-gpon-onu-mng)#ssid auth wpa wifi_0/5 key 87654321

```
OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#pon-onu-mng gpon_onu-1/1/1:1
OLT(config-gpon-onu-mng)#ssid ctrl wifi_0/1 name TREINAMENTO
OLT(config-gpon-onu-mng)#ssid auth wpa wifi_0/1 key 12345678
OLT(config-gpon-onu-mng)#ssid ctrl wifi_0/5 name TREINAMENTO5
OLT(config-gpon-onu-mng)#ssid auth wpa wifi_0/5 key 87654321
OLT(config-gpon-onu-mng)#
```

2.7. Provisionamento via PROFILE

2.7.1. LINE PROFILE

OLT#conf t

Enter configuration commands, one per line. End with CTRL/Z.

OLT(config)#pon

OLT(config-pon)#onu-profile gpon line TREINAMENTO

OLT(config-pon-gpon-line-TREINAMENTO)# vport-mode manual

OLT(config-pon-gpon-line-TREINAMENTO)# tcont 1 profile 1G

```
OLT(config-pon-gpon-line-TREINAMENTO)# gemport 1 tcont 1
OLT(config-pon-gpon-line-TREINAMENTO)# service 1 gemport 1 vlan 1000
OLT(config-pon-gpon-line-TREINAMENTO)# vport 1 map-type vlan
OLT(config-pon-gpon-line-TREINAMENTO)#
```

```
OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#pon
OLT(config-pon)#onu-profile gpon line TREINAMENTO
OLT(config-pon-gpon-line-TREINAMENTO)# vport-mode manual
OLT(config-pon-gpon-line-TREINAMENTO)# tcont 1 profile 1G
OLT(config-pon-gpon-line-TREINAMENTO)# gemport 1 tcont 1
OLT(config-pon-gpon-line-TREINAMENTO)# service 1 gemport 1 vlan 1000
OLT(config-pon-gpon-line-TREINAMENTO)# vport 1 map-type vlan
OLT(config-pon-gpon-line-TREINAMENTO)#
OLT(config-pon-gpon-line-TREINAMENTO)#
OLT#
```

2.7.2. SERVICE PROFILE

```
OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#pon
OLT(config-pon)#onu-profile gpon service PPPOE
OLT(config-pon-gpon-service-PPPOE)#wan-ip ipv4 mode pppoe username zte password zte
vlan-profile 1000 host 1
OLT(config-pon-gpon-service-PPPOE)#
```

```
OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#pon
OLT(config-pon)#onu-profile gpon service PPPOE
OLT(config-pon-gpon-service-PPPOE)#wan-ip ipv4 mode pppoe username zte password zte vlan-profile 1000 host 1
OLT(config-pon-gpon-service-PPPOE)#
```

2.7.3. Provisionamento

```
OLT(config)#interface gpon_olt-1/1/1
OLT(config-if-gpon_olt-1/1/1)#onu 1 type F6600 sn ZTEGC7879B73
OLT(config-if-gpon_olt-1/1/1)#bind-onu 1 profile line TREINAMENTO
OLT(config-if-gpon_olt-1/1/1)#bind-onu 1 profile service PPPOE
OLT(config-if-gpon_olt-1/1/1)#exit
OLT(config)#interface vport-1/1/1.1:1
OLT(config-if-vport-1/1/1.1:1)#service-port 1 user-vlan 1000 vlan 1000
OLT(config-if-vport-1/1/1.1:1)#
```

```

OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#interface gpon_olt-1/1/1
OLT(config-if-gpon_olt-1/1/1)#onu 1 type F6600 sn ZTEGC7879B73
OLT(config-if-gpon_olt-1/1/1)#bind-onu 1 profile line TREINAMENTO
OLT(config-if-gpon_olt-1/1/1)#bind-onu 1 profile service PPP0E
OLT(config-if-gpon_olt-1/1/1)#
OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#inter
OLT(config)#interface vp
OLT(config)#interface vport-1/1/1.1:1
OLT(config-if-vport-1/1/1.1:1)#ser
OLT(config-if-vport-1/1/1.1:1)#service-port 1 user-vl
OLT(config-if-vport-1/1/1.1:1)#service-port 1 user-vlan 1000 vl
OLT(config-if-vport-1/1/1.1:1)#service-port 1 user-vlan 1000 vlan 1000
OLT(config-if-vport-1/1/1.1:1)#

```

2.8. Cenário VOZ

2.8.1. Configurar Servidor VOZ

```

ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#gpon
ZXAN(config-gpon)# onu profile voip-appsrv SIP-VOIP call-waiting enable
ZXAN(config-gpon)# onu profile voip-appsrv SIP-VOIP call-transfer enable call-hold enable
ZXAN(config-gpon)# onu profile voip-media mediaprofile encoding fax-mode t38
ZXAN(config-gpon)# onu profile sip SIP-VOIP proxy 10.10.88.208 outbound 10.10.88.208
registrar 10.10.88.208
ZXAN(config-gpon)# onu profile sip SIP-VOIP rereg-time 300

```

```

ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#gpon
ZXAN(config-gpon)# onu profile voip-appsrv SIP-VOIP call-waiting enable
ZXAN(config-gpon)# onu profile voip-appsrv SIP-VOIP call-transfer enable call-hold enable
ZXAN(config-gpon)# onu profile voip-media mediaprofile encoding fax-mode t38
ZXAN(config-gpon)# onu profile sip SIP-VOIP proxy 10.10.88.208 outbound 10.10.88.208 registrar 10.10.88.208
ZXAN(config-gpon)# onu profile sip SIP-VOIP rereg-time 300
ZXAN#

```

2.8.2. Script Dados+VOZ

```

OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#interface gpon_olt-1/1/1
OLT(config-if-gpon_olt-1/1/1)#onu 1 type F6600 sn ZTEGC7879B73
OLT(config-if-gpon_olt-1/1/1)#

```

```
OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#interface gpon_olt-1/1/1
OLT(config-if-gpon_olt-1/1/1)#onu 1 type F6600 sn ZTEGC7879B73
OLT(config-if-gpon_olt-1/1/1)#
OLT#
```

```
OLT(config)#interface gpon_onu-1/1/1:1
OLT(config-if-gpon_onu-1/1/1:1)#tcont 1 profile 1G
OLT(config-if-gpon_onu-1/1/1:1)#tcont 2 profile 1G
OLT(config-if-gpon_onu-1/1/1:1)#name TREINAMENTO
OLT(config-if-gpon_onu-1/1/1:1)#vport-mode manual
OLT(config-if-gpon_onu-1/1/1:1)#gemport 1 tcont 1
OLT(config-if-gpon_onu-1/1/1:1)#gemport 2 tcont 2
OLT(config-if-gpon_onu-1/1/1:1)#vport 1 map-type vlan
OLT(config-if-gpon_onu-1/1/1:1)#vport-map 1 1 vlan 1000
OLT(config-if-gpon_onu-1/1/1:1)#vport-map 1 2 vlan 50
OLT(config-if-gpon_onu-1/1/1:1)#
```

```
OLT(config)#interface gpon_onu-1/1/1:1
OLT(config-if-gpon_onu-1/1/1:1)#tcont 1 profile 1G
OLT(config-if-gpon_onu-1/1/1:1)#tcont 2 profile 1G
OLT(config-if-gpon_onu-1/1/1:1)#name TREINAMENTO
OLT(config-if-gpon_onu-1/1/1:1)#vport-mode manual
OLT(config-if-gpon_onu-1/1/1:1)#gemport 1 tcont 1
OLT(config-if-gpon_onu-1/1/1:1)#gemport 2 tcont 2
OLT(config-if-gpon_onu-1/1/1:1)#vport 1 map-type vlan
OLT(config-if-gpon_onu-1/1/1:1)#vport-map 1 1 vlan 1000
OLT(config-if-gpon_onu-1/1/1:1)#vport-map 1 2 vlan 50
OLT(config-if-gpon_onu-1/1/1:1)#
```

```
OLT(config)#interface vport-1/1/1.1:1
OLT(config-if-vport-1/1/1.1:1)#service-port 1 user-vlan 1000 vlan 1000
OLT(config-if-vport-1/1/1.1:1)#service-port 2 user-vlan 50 vlan 50
OLT(config-if-vport-1/1/1.1:1)#
```

```
OLT(config)#interface vport-1/1/1.1:1
OLT(config-if-vport-1/1/1.1:1)#service-port 1 user-vlan 1000 vlan 1000
OLT(config-if-vport-1/1/1.1:1)#service-port 2 user-vlan 50 vlan 50
```

```
OLT(config)#pon-onu-mng gpon_onu-1/1/1:1
OLT(config-gpon-onu-mng)#service 1 gemport 1 vlan 1000
OLT(config-gpon-onu-mng)#service 2 gemport 2 vlan 50
OLT(config-gpon-onu-mng)#wan-ip ipv4 mode pppoe username f601 password f601 vlan-
profile 1000 host 1
OLT(config-gpon-onu-mng)#
OLT(config-gpon-onu-mng)#voip protocol sip
OLT(config-gpon-onu-mng)#voip-ip ipv4 mode dhcp vlan-profile 50
OLT(config-gpon-onu-mng)#sip-service pots_0/1 profile SIP-VOIP userid 200 username 200
password 200
```

```
OLT(config)#pon-onu-mng gpon_onu-1/1/1:1
OLT(config-gpon-onu-mng)#service 1 gemport 1 vlan 1000
OLT(config-gpon-onu-mng)#service 2 gemport 2 vlan 50
OLT(config-gpon-onu-mng)#wan-ip ipv4 mode pppoe username f601 password f601 vlan-profile 1000 host 1
OLT(config-gpon-onu-mng)#
OLT(config-gpon-onu-mng)#voip protocol sip
OLT(config-gpon-onu-mng)#voip-ip ipv4 mode dhcp vlan-profile 50
OLT(config-gpon-onu-mng)#sip-service pots_0/1 profile SIP-V0IP userid 200 username 200 password 200
OLT(config-gpon-onu-mng)#
```

2.9. Cenário AAA

2.9.1. Script AAA

OLT#conf t

Enter configuration commands, one per line. End with CTRL/Z.

OLT(config)#interface gpon_olt-1/1/1

OLT(config-if-gpon_olt-1/1/1)#onu 1 type F6600 sn ZTEGC7879B73

OLT(config-if-gpon_olt-1/1/1)#

```
OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#interface gpon_olt-1/1/1
OLT(config-if-gpon_olt-1/1/1)#onu 1 type F6600 sn ZTEGC7879B73
OLT(config-if-gpon_olt-1/1/1)#
OLT#
```

OLT(config)#interface gpon_onu-1/1/1:1

OLT(config-if-gpon_onu-1/1/1:1)#name TREINAMENTO

OLT(config-if-gpon_onu-1/1/1:1)#vport-mode manual

OLT(config-if-gpon_onu-1/1/1:1)#

OLT(config-if-gpon_onu-1/1/1:1)#tcont 1 profile 1G

OLT(config-if-gpon_onu-1/1/1:1)#tcont 2 profile 1G

OLT(config-if-gpon_onu-1/1/1:1)#tcont 3 profile 1G

OLT(config-if-gpon_onu-1/1/1:1)#

OLT(config-if-gpon_onu-1/1/1:1)#gemport 1 tcont 1

OLT(config-if-gpon_onu-1/1/1:1)#gemport 2 tcont 2

OLT(config-if-gpon_onu-1/1/1:1)#gemport 3 tcont 3

OLT(config-if-gpon_onu-1/1/1:1)#

OLT(config-if-gpon_onu-1/1/1:1)#vport 1 map-type vlan

OLT(config-if-gpon_onu-1/1/1:1)#vport-map 1 1 vlan 1000

OLT(config-if-gpon_onu-1/1/1:1)#vport-map 1 2 vlan 50

OLT(config-if-gpon_onu-1/1/1:1)#vport-map 1 2 vlan 1005

```
OLT(config)#interface gpon_onu-1/1/1:1
OLT(config-if-gpon_onu-1/1/1:1)#name TREINAMENTO
OLT(config-if-gpon_onu-1/1/1:1)#vport-mode manual
OLT(config-if-gpon_onu-1/1/1:1)#
OLT(config-if-gpon_onu-1/1/1:1)#tcont 1 profile 1G
OLT(config-if-gpon_onu-1/1/1:1)#tcont 2 profile 1G
OLT(config-if-gpon_onu-1/1/1:1)#tcont 3 profile 1G
OLT(config-if-gpon_onu-1/1/1:1)#
OLT(config-if-gpon_onu-1/1/1:1)#gemport 1 tcont 1
OLT(config-if-gpon_onu-1/1/1:1)#gemport 2 tcont 2
OLT(config-if-gpon_onu-1/1/1:1)#gemport 3 tcont 3
OLT(config-if-gpon_onu-1/1/1:1)#
OLT(config-if-gpon_onu-1/1/1:1)#vport 1 map-type vlan
OLT(config-if-gpon_onu-1/1/1:1)#vport-map 1 1 vlan 1000
OLT(config-if-gpon_onu-1/1/1:1)#vport-map 1 2 vlan 50
OLT(config-if-gpon_onu-1/1/1:1)#vport-map 1 2 vlan 1005
OLT(config-if-gpon_onu-1/1/1:1)#
OLT(config-if-gpon_onu-1/1/1:1)#
```

```
OLT(config)#interface vport-1/1/1.1:1
OLT(config-if-vport-1/1/1.1:1)#service-port 1 user-vlan 1000 vlan 1000
OLT(config-if-vport-1/1/1.1:1)#service-port 2 user-vlan 50 vlan 50
OLT(config-if-vport-1/1/1.1:1)#service-port 3 user-vlan 1005 vlan 1005
OLT(config-if-vport-1/1/1.1:1)#igmp fast-leave enable
```

```
OLT(config)#interface vport-1/1/1.1:1
OLT(config-if-vport-1/1/1.1:1)#service-port 1 user-vlan 1000 vlan 1000
OLT(config-if-vport-1/1/1.1:1)#service-port 2 user-vlan 50 vlan 50
OLT(config-if-vport-1/1/1.1:1)#service-port 3 user-vlan 1005 vlan 1005
OLT(config-if-vport-1/1/1.1:1)#igmp fast-leave enable
```

```
OLT(config)#pon-onu-mng gpon_onu-1/1/1:1
OLT(config-gpon-onu-mng)#service 1 gemport 1 vlan 1000
OLT(config-gpon-onu-mng)#service 2 gemport 2 vlan 50
OLT(config-gpon-onu-mng)#service 3 gemport 3 vlan 1005
OLT(config-gpon-onu-mng)#wan-ip ipv4 mode pppoe username f601 password f601 vlan-
profile 1000 host 1
OLT(config-gpon-onu-mng)#
OLT(config-gpon-onu-mng)#voip protocol sip
OLT(config-gpon-onu-mng)#voip-ip ipv4 mode dhcp vlan-profile 50
OLT(config-gpon-onu-mng)#sip-service pots_0/1 profile SIP-VOIP userid 200 username 200
password 200
OLT(config-gpon-onu-mng)#
OLT(config-gpon-onu-mng)#vlan port eth_0/4 mode tag vlan 1005
OLT(config-gpon-onu-mng)#mvlan tag eth_0/4 strip
```

```
OLT(config)#pon-onu-mng gpon_onu-1/1/1:1
OLT(config-gpon-onu-mng)#service 1 gempport 1 vlan 1000
OLT(config-gpon-onu-mng)#service 2 gempport 2 vlan 50
OLT(config-gpon-onu-mng)#service 3 gempport 3 vlan 1005
OLT(config-gpon-onu-mng)#wan-ip ipv4 mode pppoe username f601 password f601 vlan-profile 1000 host 1
OLT(config-gpon-onu-mng)#
OLT(config-gpon-onu-mng)#voip protocol sip
OLT(config-gpon-onu-mng)#voip-ip ipv4 mode dhcp vlan-profile 50
OLT(config-gpon-onu-mng)#sip-service pots_0/1 profile SIP-VOIP userid 200 username 200 password 200
OLT(config-gpon-onu-mng)#
OLT(config-gpon-onu-mng)#vlan port eth_0/4 mode tag vlan 1005
OLT(config-gpon-onu-mng)#mvlan tag eth_0/4 strip
OLT(config-gpon-onu-mng)#
```

2.10. Cenário QINQ

2.10.1. Script QINQ

```
ZXAN(config)#interface gpon_onu-1/1/3:1
ZXAN(config-if-gpon_onu-1/1/3:1)# name TESTE-ONU
ZXAN(config-if-gpon_onu-1/1/3:1)# description TESTE-ONU
ZXAN(config-if-gpon_onu-1/1/3:1)# vport-mode manual
ZXAN(config-if-gpon_onu-1/1/3:1)# tcont 1 profile 1G
ZXAN(config-if-gpon_onu-1/1/3:1)# tcont 2 profile "VoIP"
ZXAN(config-if-gpon_onu-1/1/3:1)# gempport 1 tcont 1
ZXAN(config-if-gpon_onu-1/1/3:1)# gempport 2 tcont 2
ZXAN(config-if-gpon_onu-1/1/3:1)# vport 1 map-type vlan
ZXAN(config-if-gpon_onu-1/1/3:1)# vport-map 1 1 vlan 1956
ZXAN(config-if-gpon_onu-1/1/3:1)# vport-map 1 2 vlan 3779
```

```
ZXAN(config)#interface gpon_onu-1/1/3:1
ZXAN(config-if-gpon_onu-1/1/3:1)# name TESTE-ONU
ZXAN(config-if-gpon_onu-1/1/3:1)# description TESTE-ONU
ZXAN(config-if-gpon_onu-1/1/3:1)# vport-mode manual
ZXAN(config-if-gpon_onu-1/1/3:1)# tcont 1 profile 1G
ZXAN(config-if-gpon_onu-1/1/3:1)# tcont 2 profile "VoIP"
ZXAN(config-if-gpon_onu-1/1/3:1)# gempport 1 tcont 1
ZXAN(config-if-gpon_onu-1/1/3:1)# gempport 2 tcont 2
ZXAN(config-if-gpon_onu-1/1/3:1)# vport 1 map-type vlan
ZXAN(config-if-gpon_onu-1/1/3:1)# vport-map 1 1 vlan 1956
ZXAN(config-if-gpon_onu-1/1/3:1)# vport-map 1 2 vlan 3779
ZXAN(config-if-gpon_onu-1/1/3:1)#
```

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

```
ZXAN(config)#interface vport-1/1/3.1:1
ZXAN(config-if-vport-1/1/3.1:1)#service-port 1 user-vlan 1956 vlan 1956 svlan 1047
ZXAN(config-if-vport-1/1/3.1:1)#service-port 2 user-vlan 3779 vlan 3779
```

```

ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#interface vport-1/1/3.1:1
ZXAN(config-if-vport-1/1/3.1:1)#service-port 1 user-vlan 1956 vlan 1956 svlan 1047
ZXAN(config-if-vport-1/1/3.1:1)#service-port 2 user-vlan 3779 vlan 3779
ZXAN(config-if-vport-1/1/3.1:1)#

```

```

ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#pon-onu-mng gpon_onu-1/1/3:1
ZXAN(config-gpon-onu-mng)# service 1 gempport 1 vlan 1956
ZXAN(config-gpon-onu-mng)# service 2 gempport 2 vlan 3779
ZXAN(config-gpon-onu-mng)# voip protocol sip
ZXAN(config-gpon-onu-mng)# voip-ip ipv4 mode dhcp vlan-profile VOIP host 2
ZXAN(config-gpon-onu-mng)# wan-ip ipv4 mode pppoe username teste@teste password
algar vlan-profile 1956 host 1
ZXAN(config-gpon-onu-mng)#

```

```

ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#pon-onu-mng gpon_onu-1/1/3:1
ZXAN(config-gpon-onu-mng)# service 1 gempport 1 vlan 1956
ZXAN(config-gpon-onu-mng)# service 2 gempport 2 vlan 3779
ZXAN(config-gpon-onu-mng)# voip protocol sip
ZXAN(config-gpon-onu-mng)# voip-ip ipv4 mode dhcp vlan-profile VOIP host 2
ZXAN(config-gpon-onu-mng)# wan-ip ipv4 mode pppoe username teste@teste password algar vlan-profile 1956 host 1
ZXAN(config-gpon-onu-mng)#

```

3. Atualização ONU

3.1. Atualização Manual de forma Local

3.1.1. Upload firmware para dentro da OLT

No exemplo, vamos fazer um upgrade para a versão P1T18.

```

OLT#copy ftp
//10.0.250.253/multilaser_F670L_V1.1.20P1T18_UPGRADE_BOOTLDR.bin@zte:zte root:
/datadisk0/LR0/onuver/multilaser_F670L_V1.1.20P1T18_UPGRADE_BOOTLDR.bin

```

```

OLT#copy ftp //10.0.250.253/multilaser_F670L_V1.1.20P1T18_UPGRADE_BOOTLDR.bin@zte:zte root: /datadisk0/LR0/onuver/multilaser_F670L_V1.1.20P1T18_UPGRADE_BOOTLDR.bin
Connect successfully! Start copying file
100% completed 00:00:06
Got file successfully! Received 18514076 bytes!
OLT#

```

3.1.2. Comando informando a firmware que deseja utilizar no upgrade.

```

OLT#remote-unit update-and-reboot
multilaser_F670L_V1.1.20P1T18_UPGRADE_BOOTLDR.bin gpon_olt-1/1/1 1 local

```

3.1.3. Status Update ONU

```

OLT#show remote-unit update-status gpon_olt-1/1/1 1

```

```
OLT#show remote-unit update-status gpon_olt-1/1/1 1
gpon-onu_1/1/1: 1
Taskname      : Manual
Action        : Update-and-reboot
Status        : In-progress
Progress      : 4%
Failreason    : None
Committime    :
```

Após o reboot:

```
OLT#show remote-unit update-status gpon_olt-1/1/1 1
gpon-onu_1/1/1: 1
Taskname      : Manual
Action        : Update-and-reboot
Status        : Success
Progress      : 100%
Failreason    : None
Committime    : 2021-12-28 03:52:20
```

```
OLT#show remote-unit information gpon_olt-1/1/1 1
gpon-onu_1/1/1: 1
RuVendorName  : ZTEG
RuType        : F670LV1.1.01
Region 1
Veritag       : V1.1.20P1T18
Committed     : Yes
Activated     : Yes
Valid         : Yes
Region 2
Veritag       : V1.1.20P3N4D
Committed     : No
Activated     : No
Valid         : Yes
```

3.2. Atualização via TASK

3.2.1. Firmware ONU

```
OLT#show remote-unit information gpon_olt-1/1/1 1
gpon-onu_1/1/1: 1
RuVendorName  : ZTEG
RuType        : F670LV1.1.01
Region 1
Veritag       : V1.1.20P3N4D
Committed     : Yes
Activated     : Yes
Valid         : Yes
Region 2
Veritag       : V1.1.20P3N4D
Committed     : No
Activated     : No
Valid         : Yes
```

```
OLT#show remote-unit information gpon_olt-1/1/1 2
gpon-onu_1/1/1: 2
RuVendorName : ZTEG
RuType       : F670LV1.1.01
Region 1
VerTag       : V1.1.20P1T18
Committed    : No
Activated    : No
Valid        : Yes
Region 2
VerTag       : V1.1.20P3N4D
Committed    : Yes
Activated    : Yes
Valid        : Yes
```

3.2.2. Criação da TASK

```
OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config-ru-task)# action update-and-reboot
OLT(config-ru-task)# equipment-type F670LV1.1.01
OLT(config-ru-task)# match-vertag equal V1.1.20P3N4D
OLT(config-ru-task)# mode both
OLT(config-ru-task)# location remote
OLT(config-ru-task)# parallel-limitation 10
OLT(config-ru-task)# range ne
OLT(config-ru-task)# vendor-name ZTEG
OLT(config-ru-task)# version-file
multilaser_F670L_V1.1.20P1T18_UPGRADE_BOOTLDR.bin
OLT(config-ru-task)# ipaddress 192.168.86.77 ftp
OLT(config-ru-task)# path ftp
OLT(config-ru-task)# user zte
OLT(config-ru-task)# password zte
OLT(config-ru-task)# task add
```

```
OLT(config)# remote-unit task F670L-T18
OLT(config-ru-task)# action update-and-reboot
OLT(config-ru-task)# equipment-type F670LV1.1.01
OLT(config-ru-task)# match-vertag equal V1.1.20P3N4D
OLT(config-ru-task)# mode both
OLT(config-ru-task)# location remote
OLT(config-ru-task)# parallel-limitation 10
OLT(config-ru-task)# range ne
OLT(config-ru-task)# vendor-name ZTEG
OLT(config-ru-task)# version-file multilaser_F670L_V1.1.20P1T18_UPGRADE_BOOTLDR.bin
OLT(config-ru-task)# ipaddress 192.168.86.77 ftp
OLT(config-ru-task)# path ftp
OLT(config-ru-task)# user zte
OLT(config-ru-task)# password zte
OLT(config-ru-task)# task add
OLT(config-ru-task)#
```

3.2.3. Status da TASK

```
OLT#show remote-unit summary-of task F670L-T18
Download:
file multilaser_F670L_V1.1.20P1T18_UPGRADE_BOOTLDR.bin download successful from remote server

Operating:

Waiting:
gpon-onu_1/1/1: 1-2

Fail:

Success:
```

3.2.4. Status Update ONU

```
OLT#show remote-unit update-status gpon_olt-1/1/1 1
gpon-onu_1/1/1: 1
Taskname      : F670L-T18
Action        : Update-and-reboot
Status        : Success
Progress      : 100%
Failreason    : None
Committime    : 2021-12-28 08:01:22

OLT#show remote-unit update-status gpon_olt-1/1/1 2
gpon-onu_1/1/1: 2
Taskname      : F670L-T18
Action        : Update-and-reboot
Status        : Success
Progress      : 100%
Failreason    : None
Committime    : 2021-12-28 08:01:12
```

```
OLT#show remote-unit information gpon_olt-1/1/1 1
gpon-onu_1/1/1: 1
RuVendorName : ZTEG
RuType       : F670LV1.1.01
Region 1
Veritag      : V1.1.20P3N4D
Committed    : No
Activated    : No
Valid        : Yes
Region 2
Veritag      : V1.1.20P1T18
Committed    : Yes
Activated    : Yes
Valid        : Yes

OLT#show remote-unit information gpon_olt-1/1/1 2
gpon-onu_1/1/1: 2
RuVendorName : ZTEG
RuType       : F670LV1.1.01
Region 1
Veritag      : V1.1.20P1T18
Committed    : Yes
Activated    : Yes
Valid        : Yes
Region 2
Veritag      : V1.1.20P3N4D
Committed    : No
Activated    : No
Valid        : Yes
```

3.2.5. Status Final da TASK

```
OLT#show remote-unit summary-of task F670L-T18
Download:
file multilaser_F670L_V1.1.20P1T18_UPGRADE_BOOTLDR.bin download successful from remote server

Operating:

Waiting:

Fail:

Success:
gpon-onu_1/1/1: 1-2
```

4. COMANDOS PARA TROUBLESHOOTING

4.1. ONU

4.1.1. Potência ONU

OLT#show pon power attenuation gpon_onu-1/1/1:102

```
OLT#show pon power attenuation gpon_onu-1/1/1:102
      OLT              ONU              Attenuation
-----
up    Rx : -9.264( dbm)   Tx: 2.496( dbm)      11.760( dB )
down  Tx :  5.130( dbm)   Rx: -8.001( dbm)      13.131( dB )
```

4.1.2. Detalhes ONU

OLT#show gpon onu detail-info gpon_onu-1/1/1:102

```
OLT#show gpon onu detail-info gpon_onu-1/1/1:102
```

```
ONU interface:          gpon_onu-1/1/1:102
Name:
Splitter:
Type:                   F670LV9
Configured speed mode: auto
Current speed mode:    GPON
Admin state:           enable
Phase state:           working
Config state:          success
Authentication mode:   sn
SN Bind:               enable with SN check
Serial number:         ZTEGC4A8494B
Password:
Description:
Vport mode:            gempport
DBA Mode:              Hybrid
ONU Status:            enable
OMCI BW Profile:       704kbps
OMCC Encrypt:          disable
Line Profile:          N/A
Service Profile:       N/A
ONU Distance:          170m
Online Duration:       47h 34m 09s
FEC:                   disable
FEC actual mode:       disable
1PPS+ToD:             disable
Auto replace:          disable
Multicast encryption:  disable
Multicast encryption current state: N/A
```

	Authpass Time	OfflineTime	Cause
1	2021-12-10 11:28:57	0000-00-00 00:00:00	
2	0000-00-00 00:00:00	0000-00-00 00:00:00	
3	0000-00-00 00:00:00	0000-00-00 00:00:00	
4	0000-00-00 00:00:00	0000-00-00 00:00:00	
5	0000-00-00 00:00:00	0000-00-00 00:00:00	
6	0000-00-00 00:00:00	0000-00-00 00:00:00	
7	0000-00-00 00:00:00	0000-00-00 00:00:00	
8	0000-00-00 00:00:00	0000-00-00 00:00:00	
9	0000-00-00 00:00:00	0000-00-00 00:00:00	
10	0000-00-00 00:00:00	0000-00-00 00:00:00	

4.1.3. ONUS provisionadas

```
OLT#show gpon onu state
```

```
OLT#show gpon onu state
OnuIndex      Admin state  OMCC state  Phase state  Speed mode
-----
1/1/1:1       enable      enable      working      GPON
1/1/1:2       enable      disable     DyingGasp    GPON
1/1/1:11      enable      disable     OffLine      N/A
1/1/1:12      enable      enable      working      GPON
1/1/1:20      enable      disable     LOS          GPON
1/1/1:100     enable      disable     OffLine      N/A
1/1/1:101     enable      disable     OffLine      N/A
1/1/1:102     enable      enable      working      GPON
ONU Number: 3/8
```

4.1.4. Firmware ONU

OLT#show remote-unit information gpon_olt-1/1/1 102

```
OLT#show remote-unit information gpon_olt-1/1/1 102
gpon-onu_1/1/1: 102
RuVendorName : ZTEG
RuType       : F670LV9.0
Region 1
VerTag       : V9.0.10P1N8
Committed    : No
Activated    : No
Valid        : Yes
Region 2
VerTag       : V9.0.11P1N9
Committed    : Yes
Activated    : Yes
Valid        : Yes
```

4.1.5. Status da WAN da ONU

OLT#show gpon remote-onu wan-ip gpon_onu-1/1/1:102

```
OLT#show gpon remote-onu wan-ip gpon_onu-1/1/1:102
WAN index:          1
Mode:               DHCP
ONU identifier:
Ping response:      enable
Traceroute response: disable
VLAN tag mode:      tag
CVLAN:              1005
CVLAN priority:     0
SVLAN:              0
Current IP:          192.168.83.238
Current mask:        255.255.255.0
Current gateway:     192.168.83.1
Current primary DNS: 8.8.4.4
Current secondary DNS: 0.0.0.0
Domain name:
Host name:           omci_ipv4_dhcp_1
MAC address:         5c3a.3dd1.eb63
IP host ID:          1
```

4.1.6. Status da WAN do VoIP

OLT#show gpon remote-onu voip-ip gpon_onu-1/2/1:10

```
ZXAN#show gpon remote-onu voip-ip gpon_onu-1/2/1:10
Slot index:          0
Mode:               DHCP
ONU identifier:
Ping response:      enable
Traceroute response: disable
VLAN tag mode:      tag
CVLAN:              50
CVLAN priority:     0
SVLAN:              0
Current IP:          172.16.0.12
Current mask:        255.255.255.0
Current gateway:     172.16.0.254
Current primary DNS: 10.10.88.200
Current secondary DNS: 8.8.8.8
Domain name:
Host name:           omci_ipv4_dhcp_2
MAC address:         ec6c.b5ff.42b5
IP host ID:          2
```

4.1.7. Status do Registro VoIP

OLT#show gpon remote-onu voip-linestatus gpon_onu-1/2/1:10

```
ZXAN#show gpon remote-onu voip-linestatus gpon_onu-1/2/1:10

Interface:      pots_0/1
Code used:      N/A
Service status: registered
Session type:   none
Call period 1:  0
               2:  0
Call dest address 1:0.0.0.0
                  2:0.0.0.0
Line State:     Idle, on-hook

Interface:      pots_0/2
Code used:      N/A
Service status: none
Session type:   none
Call period 1:  0
               2:  0
Call dest address 1:0.0.0.0
                  2:0.0.0.0
Line State:     Unknown or undefined
```

4.1.8. Temperatura ONU

OLT#show gpon remote-onu interface pon gpon_onu-1/1/1:102

```
OLT#show gpon remote-onu interface pon gpon_onu-1/1/1:102
Interface: pon_0/1
GEM blocklen: 48 (bytes)
SF threshold: 5
SD threshold: 9
Alarm: enable
Alarm disable interval: 0
Total T-CONT number: 8
Piggyback DBA rpt mode: mode0 only
Rx optical level: -7.102(dBm)
Lower rx optical threshold: ont internal policy
Upper rx optical threshold: ont internal policy
Tx optical level: 2.496(dBm)
Lower tx optical threshold: ont internal policy
Upper tx optical threshold: ont internal policy
ONU response time: 0(ns)
Power feed voltage: 3.24(V)
Laser bias current: 10.450(mA)
Temperature: 35.418(C)
```

4.1.9. Remover ONU

OLT#conf t

Enter configuration commands, one per line. End with CTRL/Z.

OLT(config)#interface gpon_olt-1/1/1

OLT(config-if-gpon_olt-1/1/1)#no onu 102

```
OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#inter
OLT(config)#interface gpo
OLT(config)#interface gpon_olt-1/1/1
OLT(config-if-gpon_olt-1/1/1)#no onu 102
OLT(config-if-gpon_olt-1/1/1)#
```

4.1.10. ONUS Não Provisionadas

OLT#show pon onu unc

```
OLT#show pon onu unc
OltIndex      Model      SN          PW
-----
gpon_olt-1/1/1  F670LV9.0  ZTEGC4A8494B  GC4A8494B
```

4.1.11. Distância da ONU

ZXAN#show gpon onu distance gpon_onu-1/13/1:3

```
ZXAN#show gpon onu distance gpon_onu-1/13/1:3
      Eqd (bit)      Distance(m)
.....
533590              57
```

4.1.12. ONU Retornando Sinal

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#security mac-anti-spoofing enable

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#security mac-anti-spoofing enable
ZXAN(config)#
```

4.1.13. ONU Bridge entregar DHCP

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#pon-onu-mng gpon_onu-1/13/1:3

ZXAN(config-gpon-onu-mng)#dhcp-ip ethuni eth_0/1 from-internet

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#pon-onu-mng gpon_onu-1/13/1:3
ZXAN(config-gpon-onu-mng)#dhcp-ip ethuni eth_0/1 from-internet
ZXAN(config-gpon-onu-mng)#
```

4.1.14. ONUS por PON

ZXAN#show gpon onu baseinfo gpon_olt-1/13/1

```
ZXAN#show gpon onu baseinfo gpon_olt-1/13/1
OnuIndex      Type      Mode      AuthInfo      State
-----
gpon_onu-1/13/1:3  F8648P    sn        SN:ZTEGCCDBECB8  ready
gpon_onu-1/13/1:4  F2801S    sn        SN:ZTEGCDC7242A  ready
ZXAN#
```

4.1.15. Status das ONUS

Working: A sincronização de dados está concluída. Sendo possível configurar os serviços.

```
ZTE-C620#show gpon onu state
OnuIndex      Admin state  OMCC state  Phase state  Speed mode
-----
1/1/1:1       enable      enable      working      GPON
```

Dying Gasp - A ONU está desligada e enviou o aviso antes de desligar completamente.

```
ZTE-C620#show gpon onu state
OnuIndex      Admin state  OMCC state  Phase state  Speed mode
-----
1/1/1:1       enable      disable     DyingGasp    GPON
```

LOS: A ligação entre fibra e OLT está com algum defeito.

```
ZTE-C620#show gpon onu state
OnuIndex      Admin state  OMCC state  Phase state  Speed mode
-----
1/1/1:1       enable      disable     LOS          GPON
```

Logging: A OLT encontrou a ONU e está medindo a distância.

```
ZTE-C620#show gpon onu state
OnuIndex      Admin state  OMCC state  Phase state  Speed mode
-----
1/1/1:1       enable      disable     Logging      GPON
```

SyncMib: A OLT mediu a distância até a ONU e está sincronizando os dados.

```
ZTE-C620#show gpon onu state
OnuIndex      Admin state  OMCC state  Phase state  Speed mode
-----
1/1/1:1       enable      disable     syncMib      GPON
```

Offline: A OLT não encontrou a ONU.

```
ZTE-C620#show gpon onu state
OnuIndex      Admin state  OMCC state  Phase state  Speed mode
-----
1/1/1:1       enable      disable     OffLine      N/A
```

4.1.16. Firewall ONU

ZXAN #conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN (config)#pon-onu-mng gpon_onu-1/2/1:4

ZXAN (config-gpon-onu-mng)#firewall enable level ?

high High

low Low

medium Medium

ZXAN (config-gpon-onu-mng)#firewall enable level

```

TREINAMENTO#conf t
Enter configuration commands, one per line. End with CTRL/Z.
TREINAMENTO(config)#pon-onu-mng gpon_onu-1/2/1:4
TREINAMENTO(config-gpon-onu-mng)#fire
TREINAMENTO(config-gpon-onu-mng)#firewall ?
    disable    Disable firewall function
    enable     Enable firewall function
    ip-filter  Configure parameters of IP filter
TREINAMENTO(config-gpon-onu-mng)#firewall enable level ?
    high      High
    low       Low
    medium    Medium
TREINAMENTO(config-gpon-onu-mng)#firewall enable level █

```

4.3. OLT

4.3.1. Placas instaladas no Chassi

OLT#show card

```

OLT#show card
Shelf Slot CfgType CardName      Port HardVer Status
-----
1      1      GFGH      GFGH          13  V1.0.0  INSERVICE
1      2      HFTH          16             OFFLINE
1      3      FCSDA      FCSDA          0   V1.0.0  INSERVICE
1      4      SPUF      SPUF           3   V1.0.0  INSERVICE
1      5      SPUF      SPUF           4   V1.0.0  STANDBY
1      6      PRSF      PRSF           0   V1.0.0  INSERVICE
1      7      PRSF      PRSF           0   V1.0.0  INSERVICE

```

4.3.2. Fonte de alimentação

OLT#show power

```
OLT#show power
[shelf 1 LCC]:
Total power capacity   : 3176.40W   Total used power      : 122.82W
Power type             : DC          Power group numbers   : 1
Power modules per group: 2          Division numbers     : 1
Redundancy type        : 1+1

Group 0 (N) power capacity : 0.00W
Group 0 (1) power capacity : 3176.40W
Used power               : 122.82W
```

Supply Division	Power Module	Power Name	Power Vendor	Phy Status	Power Status	Com Status	Run Status
0	0	PD700-D48A	N/A	online	normal	normal	normal
0	1	PD700-D48A	N/A	online	abnormal	normal	abnormal

Power Module	Power Temperature	Power Capacity	Output Voltage	Output Current	Power Allotted	Power Available	Software Version
0	32C	3176.40W	52.94V	2.32A	122.82W	3053.58W	V1.3
1	31C	0.00W	0.00V	0.00A	0.00W	0.00W	V1.3

4.3.3. Uptime da OLT

OLT#show software

```
OLT#show software
ZXA10 C620
ZTE ZXA10 Software, Version: V1.2.1, Release software
Copyright (c) 2020 by ZTE Corporation
Build on 2020/09/16 17:24:09
System image file is <sysdisk0: verset/C620V1.2.1.set>, file size is 346,099,451 Bytes
System image is loaded from local
System uptime is 31 day(s), 12 hour(s), 25 minute(s)

[GFGH, shelf 1, slot 1]:
Active Packages on Node PFU-1/1/0:
Base-1.2.1:
Package: Base
Version: V1.2.1

[SPUF, shelf 1, slot 4]:
Active Packages on Node MPU-1/4/0:
Base-1.2.1:
Package: Base
Version: V1.2.1

[SPUF, shelf 1, slot 5]:
Active Packages on Node MPU-1/5/0:
Base-1.2.1:
Package: Base
Version: V1.2.1
```

4.3.4. Processamento OLT

OLT#show processor

```
OLT#show processor
=====
Character: CPU current character in system
MSC      : Master-SC in Cluster System
SSC      : Slave-SC in Cluster System
N/A      : None-SC in Cluster System
CPU(5s)  : CPU usage ratio measured in 5 seconds
CPU(1m)  : CPU usage ratio measured in 1 minute
CPU(5m)  : CPU usage ratio measured in 5 minutes
Peak     : CPU peak usage ratio measured in 1 minute
PhyMem   : Physical memory (megabyte)
FreeMem  : Free memory (megabyte)
Mem      : Memory usage ratio
=====
=====
```

	Character	CPU(5s)	CPU(1m)	CPU(5m)	Peak	PhyMem	FreeMem	Mem
PFU-1/1/0	N/A	33%	33%	33%	33%	2048	742	63.770%
MPU-1/4/0	MSC	27%	27%	27%	29%	4096	2466	39.795%
MPU-1/5/0	SSC	22%	22%	22%	23%	4096	2784	32.031%

4.3.5. Detalhes do Módulo Óptico

OLT#show optical-module-info gpon_olt-1/1/1

```
OLT#show optical-module-info gpon_olt-1/1/1
Optical Module Position : gpon_olt-1/1/1
Optical Module Power State : on(No support modify)
Optical Module State : online
-----
Optical Module Manufacture Information:
-----
Vendor-Name : Hisense
Sequence-Number: U55AB006731
Product-Date : 201114
Part-Number :
Material-Number: 5a 54 45 20 30 33 33 30 33 30
31 30 30 31 36 30 00 00 00 00
Register-Data : 03 04 01 00 14 00 00 00 00 00 00 03 19 00 3c ff
00 00 00 00 48 69 73 65 6e 73 65 20 20 20 20 20
20 20 20 20 00 00 1a 95 4c 54 45 33 36 38 30 50
2d 42 43 2b 31 44 4d 20 31 2e 30 20 05 d2 00 5c
00 1c 14 14 55 35 35 41 42 30 30 36 37 33 31 20
20 20 20 20 32 30 31 31 31 34 20 20 58 e0 01 f9
5a 54 45 20 30 33 33 30 33 30 31 30 30 31 36 30
56 31 2e 30 00 00 00 00 00 00 00 00 00 00 49
-----
Optical Module Information:
-----
Module-Type : SFP/SFP+
Connector : SC
Fiber-Type : SM
OTDR : NON-OTDR
Module-Class : GPON/C+
Supply-Vol : 3.256(v)
Temperature : 28.054(c)
OTDR-Version :
```

```

-----
Optical Transmitter Information:
-----
Laser-Rate      : 25   (100Mb/s)      TxPower         : 5.131(dbm)
Wavelength      : 1490 (nm)          TxBias-Current  : 6.640(mA)
-----
Optical Module Alarm Threshold:
-----
RxPower-Upper   : 3   (dbm)           RxPower-Lower   : -20(dbm)
TxPower-Upper   : 9.000 (dbm)         TxPower-Lower   : 1.000 (dbm)
Bias-Upper      : 131(mA)             Bias-Lower      : 0(mA)
Voltage-Upper   : 7   (v)             Voltage-Lower   : 0(v)
Temperature-Upper : 90 (c)             Temperature-Lower: -45(c)
Module-Class    : default

```

4.3.6. Status das FANS

ZXAN#

ZXAN#show fan

```

ZXAN#show fan
[shelf 1 LCC]:
Fan speed mini level: 1
Group 0: online
Fan speed mode: auto
Fan      name: FAN
Index    Status      Speed
  0      work normally 28%
  1      work normally 28%
  2      work normally 28%
  3      work normally 28%

```

4.3.7. Restore OLT

OLT#copy ftp //10.0.250.253/startrun.dat@f601:f601 root: /datadisk0/DATA0/startrun.dat

```

OLT#copy ftp //10.0.250.253/startrun.dat@f601:f601 root: /datadisk0/DATA0/startrun.dat
Connect successfully! Start copying file
100% completed 00:00:00
Got file successfully! Received 35134 bytes!

```

4.3.8. Status das Portas de UPLINK

OLT#show interface brief

```

OLT#show interface brief
Interface      Portattribute Mode BW(Mbps) Admin Phy Prot Description
xgei-1/4/3     optical      Duplex/full  10000 up   up   up
xgei-1/5/1     optical      Duplex/full  10000 down down down
xgei-1/5/2     optical      Duplex/full  10000 down down down
xgei-1/5/3     optical      Duplex/full  10000 down down down
xgei-1/5/4     optical      Duplex/full  1000  down down down
smartgroup5    N/A          N/A          N/A   down up   down

```

4.3.9. MACS atrelados a VLAN

OLT#show mac vlan 1000

```
OLT#show mac vlan 1000
Total mac address : 3
```

Mac address	Vlan	Type	Port	Vc
c0b1.0137.edae	1000	Dynamic	vport-1/1/1.1:1	
5c3a.3d8a.92c1	1000	Dynamic	vport-1/1/1.12:1	
cc2d.e0c2.5158	1000	Dynamic	xgei-1/4/3	

4.3.10. Trafego de mac-address

ZXAN#show performance pon-onu mac gpon_onu-1/13/1:3 current

```
ZXAN#show performance pon-onu mac gpon_onu-1/13/1:3 current
PON-MAC-ONU gpon_onu_1/13/1:3 current performance data:
-----
PHY PM:
Rx corrected FEC bytes           :491;
Rx corrected FEC codewords       :74;
Rx uncorrectable FEC codewords   :193;
Rx total FEC codewords           :1622313892083;
Rx FS/XGTC HEC error count       :N/A;
Rx Lost Bursts                   :3623;
Rx CRC errored packets           :N/A;
Rx total bits protected by BIP    :266847387214560;
Rx BIP error count(ERRi)         :20;
Rx BER(e-9)                      :0;
DOWi count                       :N/A;

XGEM PM:
Rx XGEM frames                   :3373651672385;
Rx bytes in non-idle XGEM frames :4046304;
Rx XGEM HEC error count          :39153;
Rx FS/XGTC words lost due to XGEM HEC error :N/A;
Rx XGEM key error count          :0;
Tx XGEM frames                   :114185;
Tx bytes in non-idle XGEM frames :12898490;

PLOAM PM:
Us PLOAM message count           :406;
Us PLOAM MIC error count         :0;
Us DG count                      :0;
Us Serial_Number_ONU message count :393;
Us Registration message count     :8;
Us Key_Report message count       :8;
Us Acknowledgement message count :5;
Us Sleep_Request message count    :0;
Ds PLOAM message count           :19;
Ds Ranging_Time message count     :4;
Ds Request_Registration message count :0;
Ds Assign_Alloc-ID message count  :1;
Ds Key_Control message count      :8;
Ds Sleep_Allow message count      :0;

OMCI PM:
Us OMCI baseline message count   :1405;
Us OMCI extended message count   :0;
Us OMCI autonomous messages      :171;
Us OMCI MIC error count          :N/A;
Ds OMCI baseline message count   :1405;
Ds OMCI extended message count   :0;

ETH PM:
Rx Ethernet frames               :24462;
Rx errored Ethernet frames       :0;
ZXAN#
```

4.3.11. ALLOC ID

ZXAN#show gpon onu tcont gpon_olt-1/13/1

```
ZXAN#show gpon onu tcont gpon_olt-1/13/1
Name:
OLT:      gpon_olt-1/13/1
ONU:      3
T-CONT:   1
Alloc ID: 1024
DBA gap mode: mode2
Using following bandwidth profile:
  Profile name: 10G

Name:
OLT:      gpon_olt-1/13/1
ONU:      4
T-CONT:   1
Alloc ID: 1025
DBA gap mode: mode2
Using following bandwidth profile:
  Profile name: 10G
```

4.3.12. Remover Criptografia OLT

OLT#conf t

Enter configuration commands, one per line. End with CTRL/Z.

OLT(config)#pon

OLT(config-pon)#user-info-security disable

The configuration needs to be saved, otherwise the configuration may be lost, confirm to modify?[yes/no]:yes

OLT(config-pon)#

```
OLT#conf t
Enter configuration commands, one per line. End with CTRL/Z.
OLT(config)#pon
OLT(config-pon)#user-info-security disable
The configuration needs to be saved, otherwise the configuration may be lost, confirm to modify?[yes/no]:yes
OLT(config-pon)#
```

4.3.13. Alterar Velocidade da Porta Uplink

OLT-C610#conf t

Enter configuration commands, one per line. End with CTRL/Z.

OLT-C610(config)#interface xgei-1/1/1

OLT-C610(config-if-xgei-1/1/1)#speed

```
OLT-C610#conf t
Enter configuration commands, one per line.  End with CTRL/Z.
OLT-C610(config)#interface xgei-1/1/1
OLT-C610(config-if-xgei-1/1/1)#speed ?
    speed-100G  Eth port speed
    speed-100M  Eth port speed
    speed-10G   Eth port speed
    speed-10M   Eth port speed
    speed-1G    Eth port speed
    speed-25G   Eth port speed
    speed-40G   Eth port speed
    speed-50G   Eth port speed
```

4.3.14. LACP

```
OLT-C610#conf t
Enter configuration commands, one per line.  End with CTRL/Z.
OLT-C610(config)#interface smartgroup1
OLT-C610(config-if-smartgroup1)#show this
!<MSAN>
    switchport mode hybrid
!</MSAN>
OLT-C610(config-if-smartgroup1)#switchport mode hybrid
OLT-C610(config-if-smartgroup1)#switchport vlan 100 tag
OLT-C610(config-if-smartgroup1)#exit
OLT-C610(config)#lacp
OLT-C610(config-lacp)#lacp mode 802.3ad
OLT-C610(config-lacp)#lacp load-balance src-mac
OLT-C610(config-lacp)#exit
OLT-C610(config-lacp)#interface gei-1/1/5
OLT-C610(config-lacp-member-if-gei-1/1/5)#smartgroup 1 mode active
OLT-C610(config-lacp-member-if-gei-1/1/5)#exit
OLT-C610(config-lacp)#interface gei-1/1/6
OLT-C610(config-lacp-member-if-gei-1/1/6)#smartgroup 1 mode active
OLT-C610(config-lacp-member-if-gei-1/1/6)#exit
```

```

1. OLT-C610#conf t
2. Enter configuration commands, one per line. End with CTRL/Z.
3. OLT-C610(config)#interface smartgroup1
4. OLT-C610(config-if-smartgroup1)#show this
5. !<MSAN>
6.     switchport mode hybrid
7. !</MSAN>
8. OLT-C610(config-if-smartgroup1)#switchport mode hybrid
9. OLT-C610(config-if-smartgroup1)#switchport vlan 100 tag
10. OLT-C610(config-if-smartgroup1)#exit
11. OLT-C610(config)#lacp
12. OLT-C610(config-lacp)#lacp mode 802.3ad
13. OLT-C610(config-lacp)#lacp load-balance src-mac
14. OLT-C610(config-lacp)#exit
15. OLT-C610(config-lacp)#interface gei-1/1/5
16. OLT-C610(config-lacp-member-if-gei-1/1/5)#smartgroup 1 mode active
17. OLT-C610(config-lacp-member-if-gei-1/1/5)#exit
18. OLT-C610(config-lacp)#interface gei-1/1/6
19. OLT-C610(config-lacp-member-if-gei-1/1/6)#smartgroup 1 mode active
20. OLT-C610(config-lacp-member-if-gei-1/1/6)#exit

```

```

OLT-C610#show interface brief
Interface      Portattribute Mode BW(Mbps) Admin Phy Prot Description
gei-1/1/5      electric      Duplex/full 1000 up   up   up
gei-1/1/6      electric      Duplex/full 1000 up   up   up
xgei-1/1/1     optical       Duplex/full 1000 up   down down
xgei-1/1/2     optical       Duplex/full 1000 up   down down
xgei-1/1/3     optical       Duplex/full 1000 up   down down
xgei-1/1/4     optical       Duplex/full 1000 up   down down
smartgroup1    N/A           N/A           up   up   down

```

4.4. PON

4.4.1. Potência RX da PON

OLT#show pon power olt-rx gpon_olt-1/1/1

```

OLT#show pon power olt-rx gpon_olt-1/1/1
Onu              Rx power
-----
gpon_onu-1/1/1:1  -9.211(dbm)
gpon_onu-1/1/1:2  no signal
gpon_onu-1/1/1:11 no signal
gpon_onu-1/1/1:12 -9.354(dbm)
gpon_onu-1/1/1:20 no signal
gpon_onu-1/1/1:100 no signal
gpon_onu-1/1/1:101 no signal
gpon_onu-1/1/1:102 -9.264(dbm)

```

4.4.2. Potência TX da PON

OLT#show pon power olt-tx gpon_olt-1/1/1

```
OLT#show pon power olt-tx gpon_olt-1/1/1
Channel                Tx power
-----
1 (GPON)                5.159(dbm)
```

4.4.3. ONUS Registradas por Porta PON

OLT#show gpon onu baseinfo gpon_olt-1/1/1

```
OLT#show gpon onu baseinfo gpon_olt-1/1/1
OnuIndex                Type                Mode                AuthInfo                State
-----
gpon_onu-1/1/1:1        F601                sn                SN:ZTEGCD7FF81F        ready
gpon_onu-1/1/1:2        F6600               sn                SN:ZTEGC7879B73        ready
gpon_onu-1/1/1:11       F670L               sn                SN:ZTEGC4BA5555        ready
gpon_onu-1/1/1:12       F670L               sn                SN:ZTEGC4BAE7A2        ready
gpon_onu-1/1/1:20       F660                sn                SN:ZTEGCB01EE7         ready
gpon_onu-1/1/1:100      F670L               sn                SN:ZTEGCB01EE8         ready
gpon_onu-1/1/1:101      F670L               sn                SN:ZTEGCB01EE9         ready
gpon_onu-1/1/1:102      F670LV9             sn                SN:ZTEGC4A8494B        ready
```

4.4.4. Banda Descontada da Porta

ZXAN#show pon bandwidth dba interface gpon_olt-1/13/1

```
ZXAN#show pon bandwidth dba interface gpon_olt-1/13/1
NOTICE: If overflowflag is TRUE, it means the actual bandwidth is not enough. Please CHECK and REVISE the configuration of bandwidth!
interface                channel  configured(kbps)  free(kbps)  overflowflag  rate(x.x%)
-----
gpon_olt-1/13/1          1(XGS-PON)  0                8000000     FALSE        0.0
gpon_olt-1/13/1          1(XG-PON)   0                2000000     FALSE        0.0
gpon_olt-1/13/1          2(GPON)     0                1000000     FALSE        0.0
```

Obs.: Utilizando o commando show PON BANDWIDTH DBA INTERFACE all, é possível visualizar todas as portas de uma unica vez.

4.4.5. Copy PON

ZXAN#conf t

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#pon-copy gpon olt interface gpon_olt-1/1/2 gpon_olt-1/1/3

Please make sure to write before the operating?[yes/no]:yes

Start copy config from gpon_olt-1/1/2 to gpon_olt-1/1/3..

Copy config success!

[OK]

```
ZXAN#conf t
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#pon-copy gpon olt interface gpon_olt-1/1/2 gpon_olt-1/1/3
Please make sure to write before the operating?[yes/no]:yes
Start copy config from gpon_olt-1/1/2 to gpon_olt-1/1/3..

Copy config success!

...[OK]
```

4.5. Alarmes

4.5.1. Alarmes Ativos

OLT#show alarm current

```
OLT#show alarm current
An alarm 400152 ID 24372 level 5 occurred at 09:41:29 12-14-2021 sent by OLT MPU-1/5/0
%PORT% Physical connector works in wrong mode! Interface = xgei-1/5/3, Current work mode = media-fiber, Configured mode = media-copper
An alarm 400152 ID 24370 level 5 occurred at 09:41:10 12-14-2021 sent by OLT MPU-1/5/0
%PORT% Physical connector works in wrong mode! Interface = xgei-1/5/1, Current work mode = media-fiber, Configured mode = media-copper
An alarm 728048-(40716) ID 24368 level 2 occurred at 09:37:50 12-14-2021 sent by OLT MPU-1/4/0
%OPTICAL% Port optical module offline alarm(port:xgei-1/4/1)
An alarm 728048-(40716) ID 24367 level 2 occurred at 09:25:40 12-14-2021 sent by OLT MPU-1/4/0
%OPTICAL% Port optical module offline alarm(port:xgei-1/4/4)
An alarm 400152 ID 24302 level 5 occurred at 10:26:43 12-08-2021 sent by OLT MPU-1/4/0
%PORT% Physical connector works in wrong mode! Interface = xgei-1/4/4, Current work mode = media-fiber, Configured mode = media-copper, due to admin no shutdown
An alarm 400315 ID 33 level 3 occurred at 22:46:05 11-10-2021 sent by OLT MPU-1/4/0
%POWER% Power voltage fault! Shelf = 1, Group = A, Power-Module = 7 DC Power voltage fault alarm
An alarm 400335-(50954) ID 24 level 3 occurred at 22:45:25 11-10-2021 sent by OLT MPU-1/4/0
%POWER% Power status abnormal! Shelf = 1, Group = A, Power-Module = 7 Power status is abnormal
An alarm 400123-(33066) ID 2 level 2 occurred at 22:43:08 11-10-2021 sent by OLT MPU-1/4/0
%BOARD% Slot offline The slot = 2 in shelf = 1 is offline, boardtype is HFTH
```

4.5.2. Histórico

OLT#show alarm history

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OLT#show alarm history
An alarm 400141-(33035) ID 24516 level 3 occurred at 18:30:46 12-23-2021, cleared at 18:31:16 12-23-2021 sent by OLT MPU-1/4/0
%BOARD% The CPU utilization exceeded the prealarm threshold! (Entity = CPU, Usage = 100%, Threshold = 95%, ROUTE_MPLS[17326]:
10%,ADM[1366]:5%,FTMETHRP[1702]:5%)
An alarm 730407-(43075) ID 24502 level 5 occurred at 12:56:50 12-23-2021, cleared at 12:57:00 12-23-2021 sent by OLT PFU-1/1/0
%PON% PON alarm pon los (rack 1 shelf 1 slot 1 port 1 Probable Cause : AllOnusNeverOnline)
An alarm 730407-(43075) ID 24486 level 5 occurred at 12:55:46 12-23-2021, cleared at 12:55:54 12-23-2021 sent by OLT PFU-1/1/0
%PON% PON alarm pon los (rack 1 shelf 1 slot 1 port 1 Probable Cause : AllOnusNeverOnline)
An alarm 722445-(35113) ID 24482 level 2 occurred at 11:56:20 12-23-2021, cleared at 12:55:47 12-23-2021 sent by OLT PFU-1/1/0
%GPON% GPON alarm link olt losi(lobi) (rack 1 shelf 1 slot 1 port 1 onu 50)
An alarm 722445-(35113) ID 24481 level 2 occurred at 11:56:20 12-23-2021, cleared at 12:55:37 12-23-2021 sent by OLT PFU-1/1/0
%GPON% GPON alarm link olt losi(lobi) (rack 1 shelf 1 slot 1 port 1 onu 20)
An alarm 730408-(43076) ID 24480 level 2 occurred at 11:21:04 12-23-2021, cleared at 11:56:19 12-23-2021 sent by OLT PFU-1/1/0
%PON% PON alarm pon los (rack 1 shelf 1 slot 1 port 1 Probable Cause : LastOnuLos)
An alarm 722445-(35113) ID 24478 level 2 occurred at 11:20:25 12-23-2021, cleared at 11:21:05 12-23-2021 sent by OLT PFU-1/1/0
%GPON% GPON alarm link olt losi(lobi) (rack 1 shelf 1 slot 1 port 1 onu 50)
An alarm 722445-(35113) ID 24477 level 2 occurred at 11:20:25 12-23-2021, cleared at 11:21:05 12-23-2021 sent by OLT PFU-1/1/0
%GPON% GPON alarm link olt losi(lobi) (rack 1 shelf 1 slot 1 port 1 onu 20)
An alarm 730408-(43076) ID 24475 level 2 occurred at 11:20:06 12-23-2021, cleared at 11:20:21 12-23-2021 sent by OLT PFU-1/1/0
%PON% PON alarm pon los (rack 1 shelf 1 slot 1 port 1 Probable Cause : LastOnuLos)
An alarm 722445-(35113) ID 24473 level 2 occurred at 11:20:04 12-23-2021, cleared at 11:20:10 12-23-2021 sent by OLT PFU-1/1/0
%GPON% GPON alarm link olt losi(lobi) (rack 1 shelf 1 slot 1 port 1 onu 40)
An alarm 730407-(43075) ID 24635 level 5 occurred at 10:16:33 12-22-2021, cleared at 10:16:39 12-22-2021 sent by OLT PFU-1/1/0
%PON% PON alarm pon los (rack 1 shelf 1 slot 1 port 1 Probable Cause : AllOnusNeverOnline)
```